Characterization of *N*,*N*-Dimethylacetamide Impregnated Wood Polymer Nanocomposites (WPNCs)

M.R. Rahman

Abstract Raw wood was impregnated with *N*,*N*-dimethylacetamide to form fabricated wood polymer nanocomposites (WPNCs). FT-IR spectra showed enhanced absorption at 1419 and then 1267 cm^{-1} which confirmed the occurrence of a modification reaction. TGA data of the fabricated WPNCs indicated a better thermal stability compared to the raw wood. The dynamic Young's modulus of the WPNCs was significantly increased compared to raw wood. Through impregnation, SEM micrographs showed porous cells of raw wood was fully filled with the polymer, which led to the better stability of WPNCs. XRD analysis indicated that the crystallinity of WPNCs increased due to the increment in the stiffness as well as the thermal stability of WPNCs.

Keywords Stiffness · Modulus of elasticity (MOE) · Modulus of rupture (MOR) · Wood polymer nanocomposites (WPNCs)

1 Introduction

In the previous research, mercerization is one of the most conventional treatment methods for natural fibers. This method helps to modify its surface properties so that it can be successfully implemented in composite formation (Gassan et al. 1999). Based on the literature on the alkali treatment toward many natural fibers, the main aim of this process is to remove the hemicelluloses. However, the rate of removal of hemicellulose is depending on the concentration of the alkali used as well as the time and temperature of treatment. On the other hand, the rate of polymer impregnation is depending on the density of wood. The introduction of the polymer into the raw wood increased the density of the wood by filling the voids. This helps to improve the thermal and mechanical strength of wood as well as the

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chemical bonding of the basic macromolecules within the wood. The chemical bonding can be generally achieved by any reactive group on the removal of hydroxyl groups from the wood.

However, there is no work reported on tropical wood impregnated with *N*,*N*-dimethylacetamide polymer. This chapter presented the selected raw tropical wood species, namely *Artocarpus elasticus*, *Artocarpus rigidus, Xylopia* spp., *Koompassia malaccensis* and *Eugenia* spp., were impregnated with the aid of sodium hydroxide. Continuously, all sodium hydroxide mercerized raw woods were impregnated with *N*,*N*-dimethylacetamide after a week. This study was investigated and the outcomes were expected. The expected FT-IR showed the enhanced absorption at 1419 cm^{-1} (–C–/CH_3) and the 1267 cm^{-1} (–N–/CH_3) stretching band. This stretching band confirmed the polymerization reaction. TGA results were expected that WPNCs showed better thermal stability in comparison with the raw wood. DSC results were expected that the decomposition temperature of WPNCs provided a higher thermal stability compared to raw wood. Dynamic Young's modulus of WPNCs was expected to be significantly increased compared with the raw wood. Besides, the MOE and MOR were expected to perform better than raw wood. The Young's modulus of *Xylopia* spp., *Artocarpus rigidus,* and *Eugenia* spp. were expected to be significantly different between raw wood and WPNCs. In addition, SEM was expected to show the fully filled porous cells on the fabricated WPNCs which gave better stability of WPNCs. The increase in the stiffness and the thermal stability of WPNCs were expected to be increased due to the crystallinity of WPNCs as indicated by XRD analysis.

2 Materials and Methods

2.1 Materials

Five wood species were collected. The collected wood was divided into softwood and hardwood. The softwoods that were used, namely *Eugenia* spp., *Artocarpus rigidus*, *Artocarpus elasticus,* and *Xylopia* spp., while the hardwood chosen was *Koompassia malaccensis*. For chemicals, NaOH and *N*,*N*-Dimethylacetamid (Merck, Germany) were chosen to be used to treat the wood species. All the purity grade of the chemicals were 99%.

2.2 Manufacturing of Wood Polymer Nanocomposites

Raw wood samples were soaked into 1M sodium hydroxide solution using an autoclave for 5 h. The temperature and pressure used were 120 °C and 85 kPa,

respectively. For WPNCs, raw wood was impregnated with *N*,*N*-dimethylacetamide using a vacuum chamber at 25 °C and 60 cm Hg.

2.3 Microstructural Characterizations

2.3.1 Fourier Transform Infrared Spectroscopy (FT-IR)

Both raw wood and WPNCs fabricated were undergoing FT-IR where the infrared spectra were recorded on a Shimadzu Fourier Transform Infrared Spectroscopy (FT-IR) 81001 Spectrophotometer. The transmittance range of the scan was 4000–370 cm^{-1} for all samples.

2.3.2 X-ray Diffraction (XRD)

XRD analysis for raw wood and WPNCs were performed with a Rigaku diffractometer (Cu Kα radiation, λ = 0.1546 nm). This equipment was running at 40 kV and 30 mA. An accepted experimental procedure for the determination of degree of crystallinity was applicable to this material where most of the X-ray scattering curves were resolvable into crystalline and amorphous scattering regions.

2.3.3 Thermogravimetric Analysis (TGA)

TGA measurements of raw wood and WPNCs were carried out with about 5–10 mg of samples at heating rate of 10°C/min in a nitrogen atmosphere using a Thermogravimetric Analyzer (TA Instrument SDT Q600). All the samples were subjected to TGA in high purity nitrogen under a constant flow rate of 5 ml/min. The temperature range for the thermal decomposition of each sample occurred was from 30 to 200 °C. The weight loss and temperature of all the samples were determined and analyzed to determine the thermal degradation rate (% weight loss/min), initial degradation temperature, and residual weight at 200 °C. According to the Broido (1969), the activation energy was calculated based on the Eq. (1).

$$k = Ae^{\frac{-E_a}{RT}} \tag{1}$$

2.3.4 Differential Scanning Calorimetric (DSC)

All DSC measurements were carried out on a DSC Q10 (TA instrument) thermal system, using a sealed aluminum capsule. Each test specimen was weighed to about 3–3.5 mg, and all the samples were held at a single heating rate of 10 °C/min with scanning temperature from 30 to 200 °C. Each of the data reported represented an

average of five runs. Melting endotherm was determined which represented the integration of the area under the peak to assess the quantitative reproducibility. The crystallinity was determined and measured dynamically by integrating the peak area of a DSC curve.

2.3.5 Scanning Electron Microscopy (SEM)

Both raw wood and WPNCs were first fixed with Karnovsky's fixative. After that the samples were taken through a graded alcohol dehydration series. Once dehydrated, the specimen was coated with thin layer of gold before viewed on the SEM. The micrographs were taken at a magnification of 150, as presented in the results and discussions section.

2.3.6 Free-Free Flexural Vibration Testing

The free-free flexural vibration testing system was developed. The threads were used to support the specimen at its nodal point of specific mode. The iron plate was facing toward the electromagnet transducer and electromagnet drive where the electromagnet transducer was connected to a recorder and an oscilloscope. Besides, an electromagnet driver with the introduction of alternating current excited the vibration by attracting the iron plate at one end of specimen. The vibration produced sound that travels through the sample and detected by a microphone where the response of the microphone was connected to a sound level meter. The amplitude of sound was increased through the sound level meter, and the sound was transformed from analog signal to digital signal. All the detected signals were transferred to the PICO oscilloscope and personal computer. The frequency was varied until a maximum response was detected. The maximum frequency was named as resonant frequency. Dynamic Young's modulus (E_d) was calculated from the resonant frequency by using the Eqs. (2) and (3):

$$E' = \frac{4\pi^2 f^2 l^4 A\rho}{I(m_n)^4} \tag{2}$$

$$I = \frac{bd^3}{12} \tag{3}$$

where

d beam depth,
b beam width,
l beam length,

f natural frequency of the specimen,
ρ density,
A cross-sectional area, and

$n = 1$ is the first mode of vibration where $m_1 = 4.730$.

2.3.7 Three-Point Bending Test for MOE and MOR

The length, width, and thickness of all raw wood and WPNCs were measured. Three-point bending test was carried out using Shimadzu Universal Testing Machine with loading capacity of 300 kN and crosshead speed of 2 mm/min. MOE and MOR were carried out according to ASTM D-143 (2006). The MOE and MOR were calculated using Eqs. (4) and (5), respectively.

$$\mathrm{MOE} = \frac{L^3 m}{4bd^3} \tag{4}$$

$$\mathrm{MOR} = \frac{1.5PL}{bh^2} \tag{5}$$

where

L span length of sample, 180 mm,
b width of sample, 20 mm,
d thickness of sample, 20 mm,
m slope of the tangent to the initial line of the force displacement curve,
P the maximum breaking load, and
h depth of the beam.

2.3.8 Compression Parallel to Grain Test for Static Young's Modulus (E_s)

All the samples undergo compression parallel to grain test based on the test standard, namely ASTM D143-52. Static Young's modulus, E_s, was the only test determined using the uniaxial compression test. All the testing was carried out using Shimadzu Universal Testing Machine with loading capacity of 300kN and crosshead speed of 2 mm/min. Static Young's modulus, E_s was calculated using Eq. (6).

$$E = \frac{\text{Stress}}{\text{Strain}}$$
$$E = \frac{F/A}{\Delta L/L}$$

where

F force,
A cross-sectional area,
ΔL displacement, and
L length of sample.

$$F = \frac{EA}{L} \times \Delta L$$
$$m = \frac{EA}{L} \tag{6}$$
$$\therefore E = \frac{EA}{L}$$

3 Results and Discussion

3.1 Fourier Transform Infrared Spectroscopy (FT-IR)

Figure 1 showed the FT-IR spectra of raw wood and WPNCs. The spectrum of the raw wood showed the basic structure of all wood samples into three regions. The first region was at 4000–3300 cm^{-1} was attributed to strong broad OH stretching while the second region was attributed to C–H stretching in methyl and methylene groups at 3000–2800 cm^{-1}. The third region located from 1000 to 1750 cm^{-1} which was attributed to a strong broad superposition with sharp and discrete absorptions in the region (Owen et al. 1989). There were few peaks to detect for hemicellulose and lignin. The peaks at 1508 and 1600 cm^{-1} were caused by lignin. For lignin, it could be detected at peak of 1734 cm^{-1}. According to Owen et al. (1989), the C=O stretch in non-conjugated ketones, carbonyls, and in ester groups was investigated. Moreover, the region between 1800 and 1100 cm^{-1} was compared with the bands assigned to the main components from wood such as cellulose, hemicelluloses, and lignin. From Fig. 1, it was clearly detected that both raw wood and WPNCs showed different infrared spectra and location. The reaction between raw wood and *N*,*N*-dimethylacetamide was well defined and was shown in Fig. 2.

The peak at 3424 cm^{-1} was attributed to the intensity of –OH absorption band. This peak was greatly reduced on WPNCs after the impregnation of *N*,*N*-dimethylacetamide into raw wood. However, the peak at 1734 cm^{-1} was clearly observed in raw wood which represented the higher xylan content in raw wood. For polymerized wood, this peak was clearly shifted to 1570 cm^{-1} due to the stronger carbonyl bands within the polymerized wood. In addition, the absorption at 1419 and 1267 cm^{-1} was attributed to –C–/CH_3 and –N–/CH_3 stretching band which confirmed the polymerization reaction.

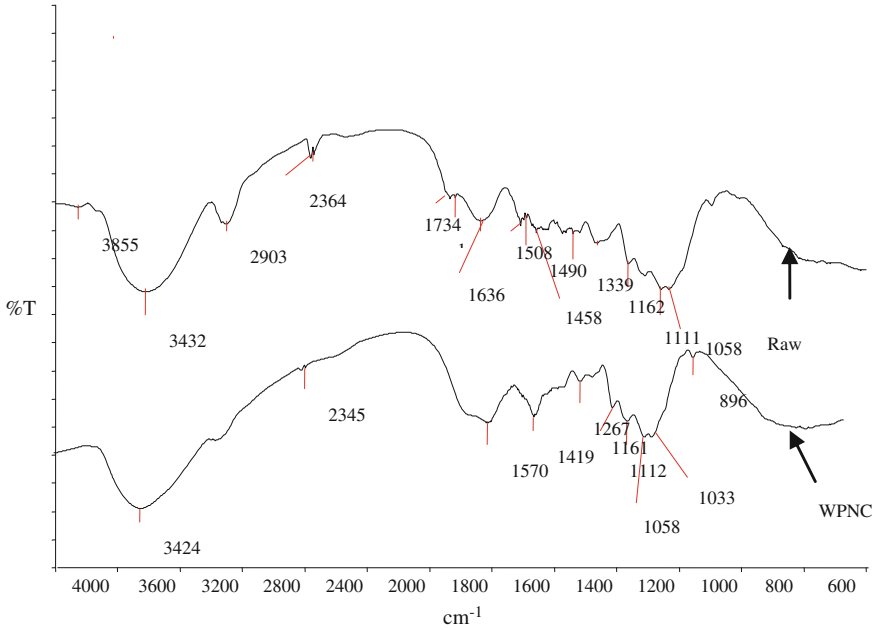
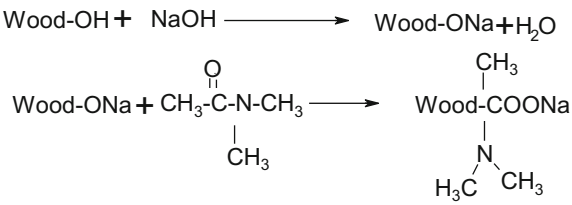


Fig. 1 IR spectrum of raw wood and WPNCs

Fig. 2 Mecerization of wood specimens with polymerization reaction



3.2 *Thermogravimetric Analysis (TGA)*

Thermogravimetric Analysis (TGA) was used to determine and investigate the thermal stability of wood. TGA is a very important parameter in the production of WPNCs. According to Wielage et al. (1999), wood acted as a filler material to improve mechanical strength in plastics that were processed at temperature below 200 °C. Through the TGA results, raw wood undergone dehydration process where there was 5–8% of absorbed water was removed. From the literature, it had been established that there was no degradation up to 160 °C (Wielage et al. 1999). Above this temperature, thermal stability of WPNCs on different species gradually decreased and the decomposition took place. Both Fig. 3a–e and Table 1 represented the TGA curves obtained by the dynamic scans for raw wood and WPNCs.

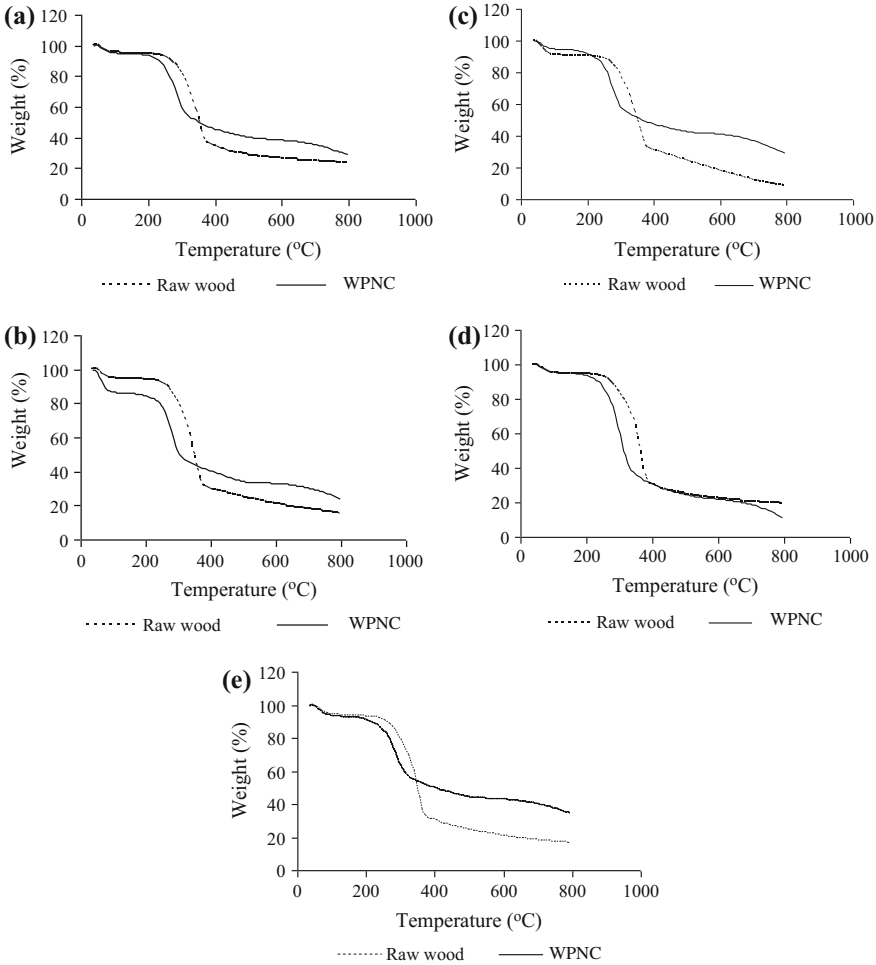


Fig. 3 **a** TGA curve of *Artocarpus elasticus* raw wood and WPNCs, **b** TGA curve of *Artocarpus rigidus* raw wood and WPNCs, **c** TGA curve of *Xylopia* spp. raw wood and WPNCs, **d** TGA curve of *Koompassia malaccensis* raw wood and WPNCs, **e** TGA curve of *Eugenia* spp. raw wood and WPNCs

The TGA curves of raw wood and WPNCs indicated that water started at 10 °C till 150 °C for both raw and WPNCs. *N*,*N*-dimethylacetamide was well impregnated into WPNCs to improve the thermal stability of wood. Besides, the evidence that the initial and final decomposition temperature (T_i, T_f) of all WPNCs was higher than raw wood. Therefore, the activation energy was calculated which could be helpful in reaching conclusions on the thermal stability of WPNCs. From the analysis, higher activation energy represented higher thermal stability.

Table 1 Thermal characteristics of raw wood and WPNCs

Sample		T_i (°C)	T_m (°C)	T_f (°C)	W_i (%)	W_m (%)	W_f (%)	Activation energy, E_a (J/K)
Artocarpus elasticus	Raw wood	201	332	419	98.96	64.73	32.67	33.11
	WPNCs	210	269	622	97.89	77.16	38.17	73.49
Artocarpus rigidus	Raw wood	211	340	388	99.67	56.45	30.63	25.79
	WPNCs	215	277	537	98.16	63.46	33.68	34.99
Xylopia spp.	Raw wood	212	341	420	98.78	56.15	29.85	27.04
	WPNCs	225	285	630	98.38	64.35	41.16	30.14
Koompassia malaccensis	Raw wood	212	355	403	98.99	57.91	30.00	16.12
	WPNCs	213	301	576	98.06	59.94	22.51	41.34
Eugenia spp.	Raw wood	214	340	443	99.77	60.32	27.51	19.17
	WPNCs	218	285	521	98.48	70.68	44.35	28.68

Arrhenius equation was used to measure the TGA thermograms of all raw wood and WPNCs based on Broido's research (1973) as shown in Eq. (1). The activation energy could be obtained from the plotting of ln k versus $1/T$. The linear plots were shown in Fig. 4a–e where the value for activation energy was tabulated in Table 1. The activation energy of WPNCs was higher compared with raw wood on different types of wood species.

3.3 Differential Scanning Calorimetry (DSC)

Thermogravimetric Analysis was carried out to determine thermal analysis as well as to determine the moisture content and volatile components presented in WPNCs. According to Herrera-France and Aguilar-Vega (1997), the properties of WPNCs were deteriorated due to the moisture content and volatile components that reduced the application of WPNCs. Through DSC analysis, raw wood and WPNCs were investigated and the results were analyzed. The thermal behavior of raw wood and WPNCs was determined and analyzed. In addition, DSC analysis also enabled the chemical activity within the raw wood and WPNCs over the temperature.

The DSC curves of raw wood and WPNCs were presented in Fig. 5a–e. Besides, Table 2 presented the enthalpy, and exotherm peaks were analyzed. Both raw wood and WPNCs showed a broad endotherm observed in the temperature range approximately 50–161 °C for all species. During this temperature range, water molecules were presented within the fibers. Based on the first decomposition temperature, WPNCs provided a higher value than the raw wood which defined the

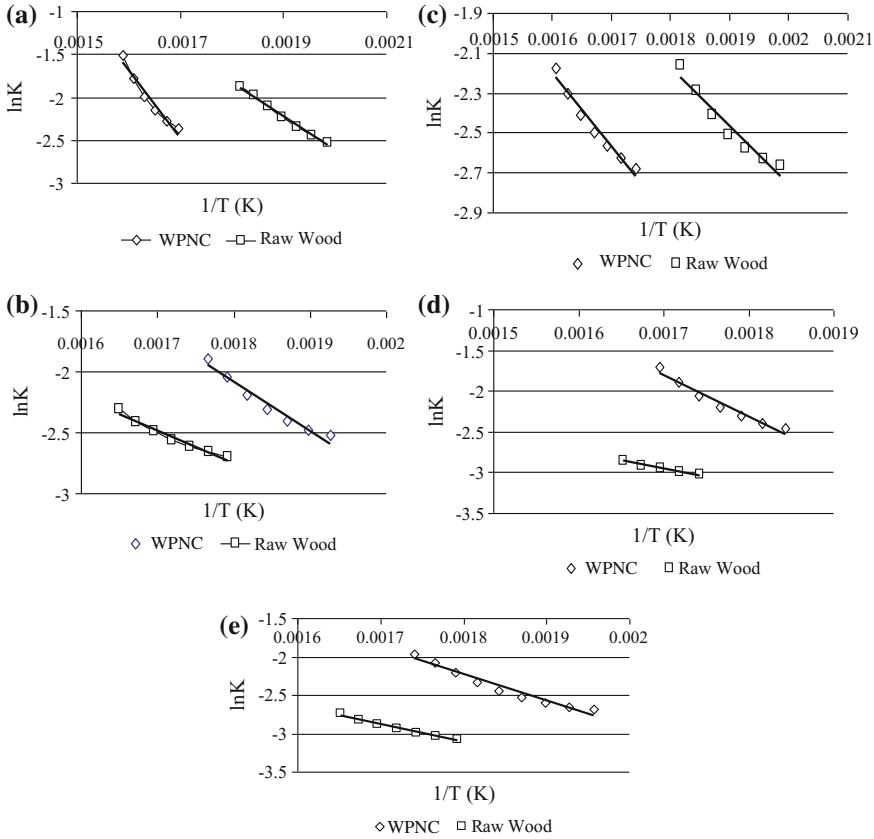


Fig. 4 **a** Temperature dependency of the decomposition rate of the *Artocarpus elasticus* raw wood and WPNCs, **b** Temperature dependency of the decomposition rate of the *Artocarpus rigidus* raw wood and WPNCs, **c** Temperature dependency of the decomposition rate of the *Xylopia* spp. raw wood and WPNCs, **d** Temperature dependency of the decomposition rate of the *Koompassia malaccensis* raw wood and WPNCs, **e** Temperature dependency of the decomposition rate of the *Eugenia* spp. raw wood and WPNCs

higher thermal stability within WPNCs. This was due to the bond between the functional groups within *N*,*N*-dimethylacetamide and the wood carboxyl. Based on the research work by Akita and Kase (1967), lignin degraded at a temperature around 200 °C within the cellulose fibers while the other polysaccharides such as cellulose performed thermal degradation at higher temperature. Thus, the second endothermic peaks which were higher than 200 °C, represented the decomposition temperatures of the cellulose in the wood fibers. There were two main decompositions namely the first thermal decomposition and second thermal decomposition. In addition, alkali and silane treatment increased the thermal stability of WPNCs

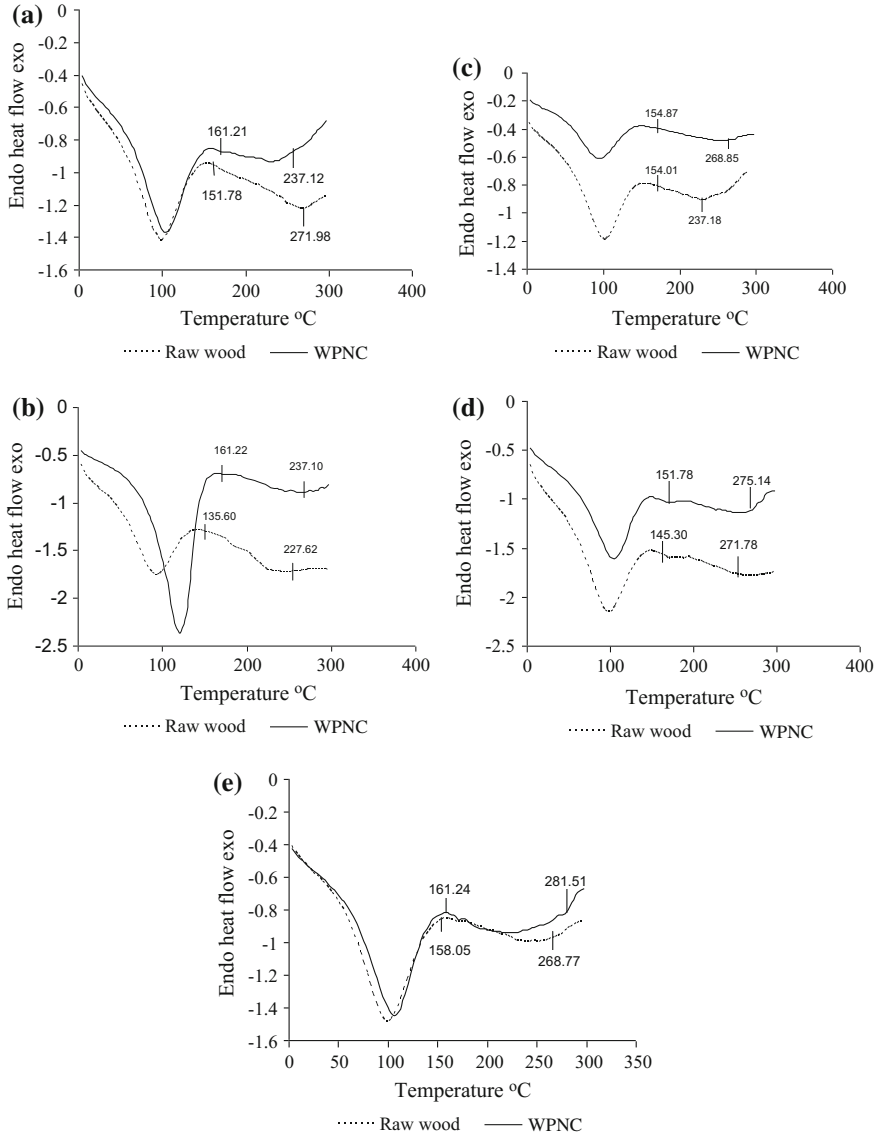


Fig. 5 **a** DSC thermographs of the *Artocarpus elasticus* raw wood and WPNCs, **b** DSC thermographs of the *Artocarpus rigidus* raw wood and WPNCs, **c** DSC thermographs of the *Koompasia malacennis* raw wood and WPNCs, **d** DSC thermographs of the *Xylopia* spp. raw wood and WPNCs, **e** DSC thermographs of the *Eugennia* spp. raw wood and WPNCs

Table 2 Crystallization enthalpy and exotherm peaks of raw wood and WPNCs

Species	Crystallization enthalpy ΔH_c (J/gm)		1st exotherm peaks (°C)	
	Raw wood	WPNCs	Raw wood	WPNCs
Artocarpus elasticus	232.10	238.64	151.78	161.21
Artocarpus rigidus	248.64	474.16	135.60	161.22
Koompasia malacennis	192.86	194.20	154.01	154.87
Xylopia spp.	252.31	323.00	145.30	151.78
Eugennia spp.	248.20	288.78	158.05	161.24

(Sreekala et al. 1997). Table 2 showed the crystallization enthalpy (ΔH_c) of all WPNCs higher than raw wood that indicated that WPNCs were more crystalline compared with raw wood, as reflected in the XRD results.

3.4 Dynamic Young's Modulus

Free-free flexural vibration testing system provided the stiffness of the raw wood and WPNCs as shown in Fig. 6. From this characterization, ten specimens were used for each wood species to ensure the accuracy of the results. The N,N-dimethylacetamide impregnation increased the Young's modulus of WPNCs on different wood species. Activation energy was significantly affected by the polymerization for WPNCs as shown in Table 1. However, only WPNCs *Artocarpus elasticus* showed significant changes in the elastic properties among all the WPNCs as well as raw wood.

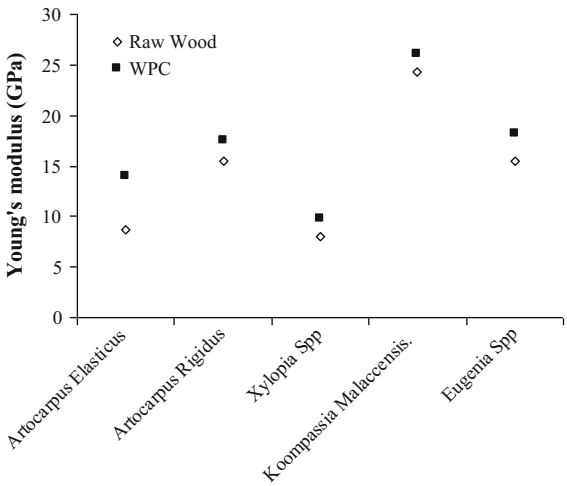


Fig. 6 E_d of raw wood and WPNCs for all species

3.5 MOE and MOR Measurement

Tables 3 and 4 as well as Figs. 7 and 8 showed MOE and MOR results for raw wood and WPNCs on different types of wood species through the impregnation of *N*,*N*-dimethylacetamide. WPNCs on *Artocarpus elasticus* and *Artocarpus rigidus* showed higher MOE values while WPNCs on *Eugenia* spp., *Xylopia* spp., and *Koompassia malaccensis* showed moderate MOE values, respectively. From Fig. 7, it was clearly shown that WPNCs yielded higher MOE compared to the raw wood due to the *N*,*N*-dimethylacetamide impregnation, which was agreed by other research (Adams et al. 1970; Yildiz et al. 2005).

Table 3 showed that the increment of MOE on *Artocarpus elasticus*, *Artocarpus rigidus,* and *Eugenia* spp. were significantly higher than raw wood. However, the

Table 3 *t*-test analysis of raw wood and wood polymer nanocomposites[a]

Treatment	Modulus of elasticity	*t*-test grouping[b]
Raw wood (*Artocarpus elasticus*)	6.48 ± 0.52	A
WPNCs (*Artocarpus elasticus*)	12.44 ± 0.99	B
Raw wood (*Artocarpus rigidus*)	5.13 ± 0.38	C
WPNCs (*Artocarpus rigidus*)	9.56 ± 1.97	D
Raw wood (*Xylopia* spp.)	6.71 ± 0.34	E
WPNCs (*Xylopia* spp.)	9.23 ± 1.81	E
Raw wood (*Koompassia malaccensis*)	15.68 ± 1.11	F
WPNCs (*Koompassia malaccensis*)	16.75 ± 0.96	F
Raw wood (*Eugenia* spp.)	9.25 ± 0.37	G
WPNCs (*Eugenia* spp.)	13.16 ± 0.41	H

[a]Each value is the average of 10 specimens
[b]The same letters are not significantly different at $\alpha = 5\%$

Table 4 *t*-test analysis of raw wood and wood polymer nanocomposites[a]

Treatment	Modulus of elasticity	*t*-test grouping[b]
Raw wood (*Artocarpus elasticus*)	76.17 ± 3.01	A
WPNCs (*Artocarpus elasticus*)	83.92 ± 4.20	B
Raw wood (*Artocarpus rigidus*)	33.02 ± 3.63	C
WPNCs (*Artocarpus rigidus*)	73.05 ± 9.26	D
Raw wood (*Xylopia* spp.)	45.91 ± 1.36	E
WPNCs (*Xylopia* spp.)	80.35 ± 9.01	E
Raw wood (*Koompassia malaccensis*)	122.20 ± 15.49	F
WPNCs (*Koompassia malaccensis*)	125.98 ± 6.77	F
Raw wood (*Eugenia* spp.)	46.10 ± 3.62	G
WPNCs (*Eugenia* spp.)	72.8 ± 4.60	H

[a]Each value is the average of 10 specimens
[b]The same letters are not significantly different at $\alpha = 5\%$

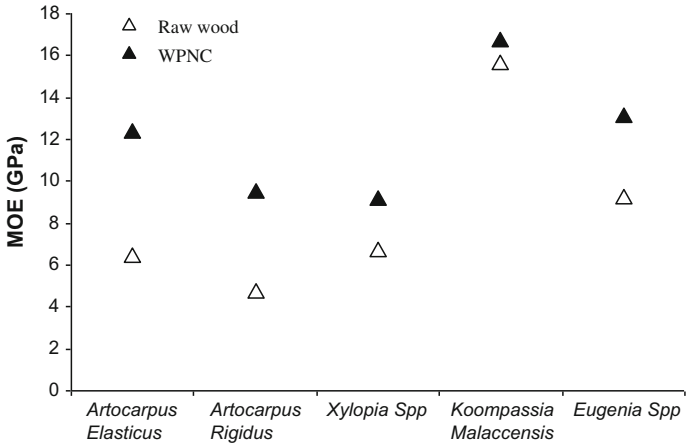


Fig. 7 MOE of raw wood and WPNCs

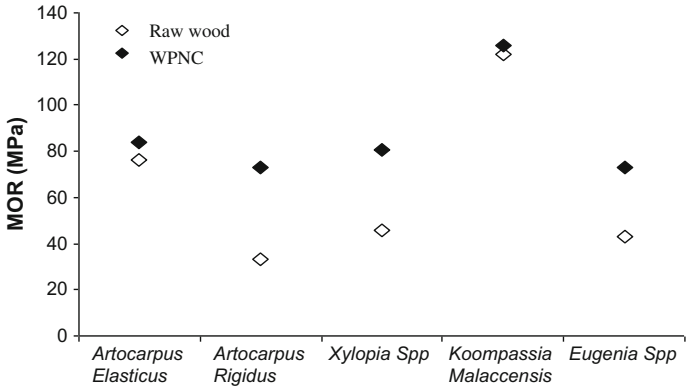


Fig. 8 MOR of raw wood and WPNCs

only hardwood *Koompassia malaccensis* as well as softwood *Xylopia* spp. showed less significant effect on the impregnation of *N*,*N*-dimethylacetamide as they had better hardness in origin form.

In this study, sodium hydroxide reacted with the cellulose in wood cells. This reaction reduced the water molecules from the raw wood to fabricate better WPNCs. *N*,*N*-Dimethylacetamid impregnation on raw wood managed to fill the void spaces completely to increase its stiffness.

MOR of WPNCs were increased with the impregnation of *N*,*N*-dimethylacetamide in agreement with previous research (Adams et al. 1970). Besides, MOR of WPNCs on *Artocarpus rigidus*, *Xylopia* spp., and *Eugenia* spp. were significantly improved as tabulated in Table 4. Figure 8 showed that MOR for *Artocarpus rigidus* was the highest while *Xylopia* spp., *Eugenia* spp., *Artocarpus elasticus, and*

Koompassia malaccensis showed moderate MOR values. Both raw wood and WPNCs hardwood *Koompassia malaccensis* showed almost similar values which indicated that *N*,*N*-dimethylacetamide impregnation was not effective on hardwood which proven the statement on our previous work (Rahman et al. 2010).

3.6 Static Young's Modulus (E)

Figure 9 showed the static Young's modulus was determined from 10 repetitions. WPNCs on *Xylopia* spp. showed the highest increment of *E* value, followed by *Artocarpus rigidus*, *Eugenia* spp., *Artocarpus elasticus,* and *Koompassia malaccensis*, respectively. The increment of Young's modulus for WPNCs on *Xylopia* spp., *Artocarpus rigidus,* and *Eugenia* spp. was significantly higher than raw wood. The increment of *E* in WPNCs was also reported by different researchers (Autio et al. 1970; Hamdan et al. 2010). With the introduction of *N*,*N*-dimethylacetamide into WPNCs, it acted as plasticizer on the cell walls through thickening which gradually improved the lateral stability (Table 5).

3.7 X-ray Diffraction (XRD)

Figures 10 and 11 showed the X-ray diffraction patterns of raw wood and WPNCs. The patterns of raw wood exhibited only one well-defined 2θ peak at 22.0° as shown in Fig. 10, which was due to cellulose (Yi et al. 2010). On the other hand, WPNCs exhibited four broad 2θ peaks at 42.08°, 43.68°, 49.23°, and 72.67°. All these peaks were observed due to the introduction of polymer into raw wood that caused polymerization. According to the report of Yi et al. (2010), chemical

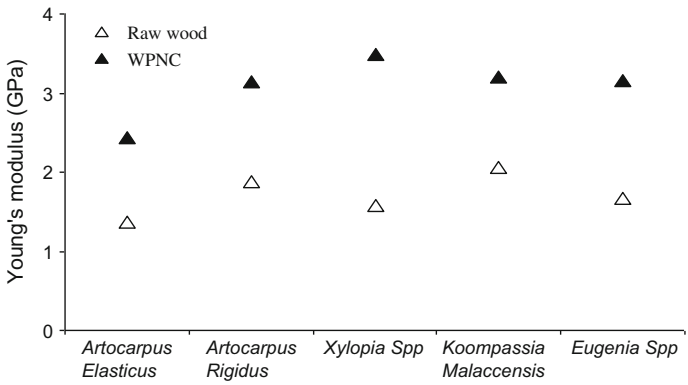


Fig. 9 Static Young's modulus of raw wood and WPNCs

Table 5 *t*-test analysis of raw wood and wood polymer nanocomposites[a]

Treatment	Static young's modulus	*t*-test grouping[b]
Raw wood (*Artocarpus elasticus*)	1.36 ± 0.23	A
WPNCs (*Artocarpus elasticus*)	2.43 ± 0.51	B
Raw wood (*Artocarpus rigidus*)	1.88 ± 0.58	C
WPNCs (*Artocarpus rigidus*)	3.14 ± 1.11	D
Raw wood (*Xylopia* spp.)	1.57 ± 0.39	E
WPNCs (*Xylopia* spp.)	3.48 ± 0.37	E
Raw wood (*Koompassia malaccensis*)	2.05 ± 0.18	F
WPNCs (*Koompassia malaccensis*)	3.19 ± 0.31	F
Raw wood (*Eugenia* spp.)	1.67 ± 0.60	G
WPNCs (*Eugenia* spp.)	3.16 ± 0.75	H

[a]Each value is the average of 10 specimens
[b]The same letters are not significantly different at $\alpha = 5\%$

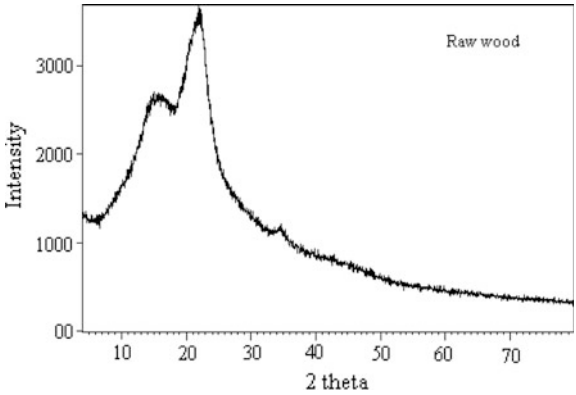


Fig. 10 X-ray diffraction of raw wood fiber

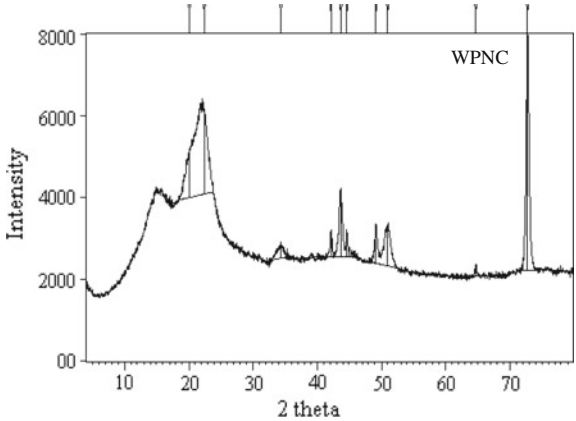


Fig. 11 X-ray diffraction of WPNCs

modification of lignocellulosic materials greatly affected their crystallinity. The impregnation process started with the reagent reacted with the chain ended on the surface of crystallites. This occurred as the polymers could not diffuse into the crystalline region. This resulted in the opening of some of the hydrogen-bonded cellulose chains. Through this, it was significantly proved that the impregnation of *N*,*N*-dimethylacetamide into raw wood helped to increase the crystallinity of WPNCs, by an drastic removal of the fibers' amorphous constituents.

3.8 Scanning Electron Microscopy (SEM)

SEM micrograph for the raw wood and WPNCs were clearly presented in Fig. 12. Figure 12a showed the SEM micrograph of raw wood on *Artocarpus rigidus* while Fig. 12b showed the SEM micrograph of WPNCs. The changes between Fig. 12a,

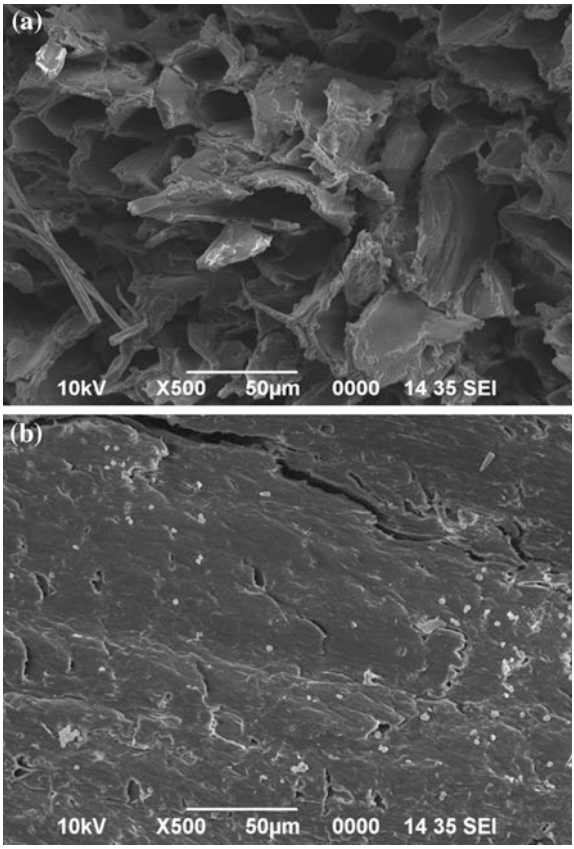


Fig. 12 SEM micrograph of raw wood (**a**) and WPNCs (**b**)

b were dramatic which proved the changes after the impregnation process. The dramatic changes on the surface morphology would led to the drastic increase of activation energy as well as the final decomposition temperature of WPNC after polymerization as shown in Fig. 3a–e and Table 1. The surface morphology proved that raw wood consisted of many void spaces within the wood cell wall itself. However, all the void spaces were fully covered by *N*,*N*-diemthylacetamide through impregnation process. At the same time, waxy substances were removed from fabricated WPNCs which was confirmed by our previous work (Rahman et al. 2010).

4 Conclusion

The present study was to investigate the introduction of *N*,*N*-dimethylacetamide into raw wood. FT-IR spectra indicated the reaction between organic *N*,*N*-Dimethylacetamid and raw wood through impregnation. The absorption band at 1419 cm^{-1} which represented –C–/CH_3 and 1267 cm^{-1} was attributed to –N–/CH_3 stretching bands confirmed the polymerization reaction. This polymerization of *N*, *N*-dimethylacetamide with raw wood improved WPNCs thermally over certain temperature range compared to raw wood. Therefore, WPNCs showed higher thermal stability among all the samples prepared. Through mechanical testing, the stiffness, MOE, and MOR of the WPNCs was significantly increased compared with raw wood on *Artocarpus elasticus, Artocarpus rigidus,* and *Eugenia* spp., respectively. In addition, the Young's modulus of *Xylopia* spp., *Artocarpus rigidus,* and *Eugenia* spp. was significantly different compared to raw wood and other WPNCs. X-ray diffraction patterns indicated WPNCs were more crystalline compared with raw wood.

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